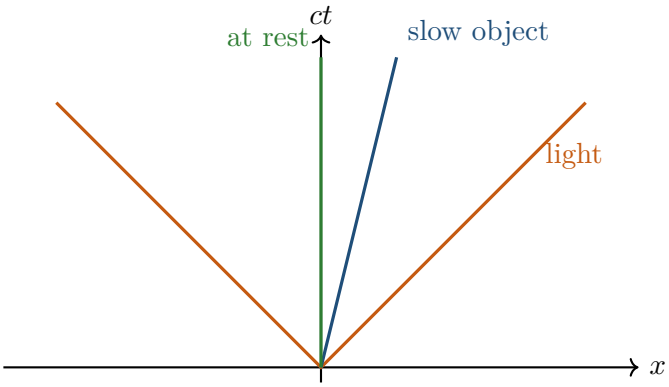
Relativity from Zero for Computer Science Students

Lecture Notes for Absolute Beginners

Prepared as self-study lecture notes



A first spacetime diagram: time upward, space horizontally

Key idea

These notes are written for students who know very little physics and may feel insecure with mathematics. The goal is not to memorize formulas. The goal is to understand why relativity was needed, what the formulas mean, and how to use them without panic.

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1 How to Read These Notes

Relativity has a reputation for being mysterious. Most of that mystery comes from two problems:

1. People try to learn the formulas before they understand the meaning of measurement.
2. People mix everyday intuition with situations involving speeds close to the speed of light.

In ordinary life, speeds are tiny compared with the speed of light. Because of that, Newtonian mechanics feels obvious. Relativity becomes visible only when velocities are a significant fraction of the speed of light, or when gravitational fields are very strong, or when measurements must be extremely precise.

For computer science students, relativity is also a useful lesson in modeling. The same physical event can have different coordinate descriptions, just like the same data structure can have different representations. The representation changes; the underlying object does not.

Key idea

Relativity is not the statement that "everything is subjective". It is the statement that different observers may assign different coordinates to the same events, while certain quantities remain invariant.

2 Mini Mathematical Survival Kit

This course uses only a few mathematical ideas. We list them here because many mistakes in relativity are not physics mistakes but notation mistakes.

2.1 Fractions and square roots

The most common expression in special relativity is

$$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}}.$$

Read it slowly:

1. Divide the speed by the speed of light: v/c.
2. Square the result: $(v/c)^2$.
3. Subtract from 1.
4. Take the square root.
5. Take the reciprocal.

Worked example

For $v = 0.8c$:

$$\frac{v}{c} = 0.8, \quad \left(\frac{v}{c}\right)^2 = 0.64, \quad 1 - 0.64 = 0.36, \quad \sqrt{0.36} = 0.6, \quad \gamma = \frac{1}{0.6} = 1.667.$$

2.2 Units

The speed of light is approximately

$$c = 3.00 \times 10^8 \,\mathrm{m/s}.$$

A distance such as $c \cdot 1\,\mathrm{s}$ means the distance light travels in one second:

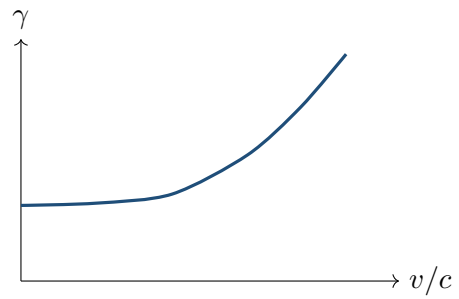
$$c \cdot 1\,\mathrm{s} = 3.00 \times 10^8 \,\mathrm{m}.$$

Common mistake

Do not add meters and seconds. Expressions like $x + ct$ are meaningful because ct has units of length. The factor c converts time into a length scale.

2.3 Graphs

A graph is not a decoration. In relativity, a graph often tells us what is possible and what is impossible.



as speed approaches c, γ grows quickly

2.4 Approximation

If $v \ll c$, then v^2/c^2 is extremely small. For example, if $v = 300\,\mathrm{m/s}$, then

$$\frac{v}{c} = \frac{300}{3 \times 10^8} = 10^{-6}, \qquad \frac{v^2}{c^2} = 10^{-12}.$$

So

$$\gamma \approx 1.$$

This is why Newtonian mechanics works for cars, balls, trains, and everyday projectiles.

Key idea

Relativity does not destroy old physics. It explains why old physics works as an approximation at low speeds and weak gravity.

3 Before Relativity: The Classical Picture

3.1 Events, objects, and coordinates

An **object** is something that exists over time: a ball, a spaceship, a detector, a clock. An **event** is something that happens at one place and one time: a flash of light, a collision, a clock tick, a

signal arriving.

In one spatial dimension an event is described by two numbers:

$$(t, x).$$

In three spatial dimensions it is described by four numbers:

$$(t, x, y, z).$$

Definition

An **event** is a point in spacetime. In special relativity, an event is usually labeled by four coordinates:

$$(t, x, y, z).$$

The event itself is physical. The coordinates are the description used by a particular observer.

This distinction is essential. A database row and its displayed formatting are not the same thing. Similarly, an event and its coordinates are not the same thing.

3.2 Newtonian assumptions

In Newtonian mechanics, the following assumptions are taken for granted:

1. Time is universal. All observers agree on time intervals.
2. Space is universal. All observers agree on lengths.
3. Velocities add in the obvious way.
4. There is no maximum possible speed.

If a train moves at speed v and a passenger walks forward at speed u relative to the train, a person on the platform says that the passenger moves at

$$u + v.$$

This is the ordinary velocity addition rule.

Common mistake

Newtonian mechanics is not stupid or wrong in everyday conditions. It is an approximation that works extremely well when all speeds are much smaller than the speed of light.

3.3 Galilean transformations

Suppose observer S' moves with speed v along the x-axis relative to observer S. At time $t = 0$, their origins coincide. In Newtonian mechanics,

$$x' = x - vt, \qquad t' = t.$$

The second equation is the crucial one: both observers use the same time.

Worked example

A car moves at 30 m/s relative to the road. A ball inside the car moves forward at 5 m/s relative to the car. According to the road observer, the ball moves at

$$30 + 5 = 35\,\mathrm{m/s}.$$

This is ordinary Galilean velocity addition.

3.4 Why the classical picture looked convincing

For centuries this picture worked beautifully. It explained projectiles, planets, pendulums, machines, and fluids. The problem appeared when physicists understood light more deeply.

4 The Speed of Light Problem

4.1 Light as an electromagnetic wave

Maxwell's equations predicted electromagnetic waves moving with speed

$$c = \frac{1}{\sqrt{\varepsilon_0 \mu_0}}.$$

Experimentally this speed is approximately

$$c \approx 3.00 \times 10^8\,\mathrm{m/s}.$$

This number is not just large. It is structurally important. It appears in the equations of electromagnetism themselves.

Key idea

In special relativity, c is not merely "the speed of light". It is the invariant speed of spacetime: the maximum speed at which causal influence can propagate.

4.2 The conflict

If velocities simply add, then a person moving toward a light beam should measure a larger speed of light, and a person moving away should measure a smaller speed. But Maxwell's theory seemed to give one fixed speed.

The conflict was:

Newtonian expectation	Maxwell/electromagnetic prediction
Light speed should depend on observer motion.	Light speed appears built into the laws of electromagnetism.
Time is universal.	Electromagnetic waves suggest a deeper spacetime structure.
Velocities add normally.	No observer should measure a different c.

4.3 Michelson-Morley in one paragraph

The Michelson-Morley experiment tried to detect Earth's motion through a hypothetical medium called the ether. If light needed an ether, Earth moving through the ether should affect the measured speed of light in different directions. The expected effect was not found.

We do not need the experimental details here. The conceptual message is enough:

Key idea

The speed of light did not behave like the speed of an ordinary object moving through an ordinary medium.

5 Einstein's Two Postulates

In 1905, Einstein reorganized the problem around two postulates.

Definition

Postulate 1: Principle of relativity. The laws of physics are the same in all inertial frames.
Postulate 2: Constancy of the speed of light. Every inertial observer measures the same speed c for light in vacuum.

An **inertial frame** is a frame that is not accelerating. Roughly: a coordinate system attached to an observer moving at constant velocity.

5.1 The shocking consequence

The two postulates cannot coexist with universal time. Something must change.

If all inertial observers measure the same speed of light, then observers in relative motion cannot all keep the same clock synchronization, same time intervals, and same lengths. Time and space must mix.

Common mistake

Relativity does not say that measurements are random. It says that measurements depend on the observer's frame according to precise equations.

6 Spacetime Diagrams for Beginners

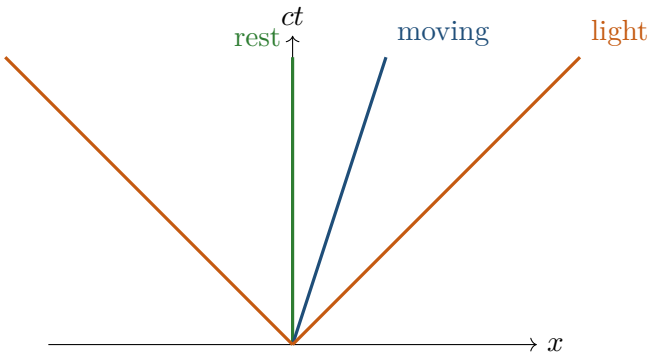
6.1 Why draw time upward?

A spacetime diagram uses one axis for space and one axis for time. Usually we draw x horizontally and ct vertically. Multiplying time by c makes both axes have units of length.

A point in this diagram is an event. A curve in this diagram is a worldline.

Definition

A **worldline** is the path of an object through spacetime. It is not an ordinary path through space. It shows where the object is at each time.



Worldlines in a spacetime diagram

6.2 Reading slope as speed

In an ordinary position-time graph, speed is slope $\Delta x/\Delta t$. In a spacetime diagram with ct vertical, the visual slope is inverted. A vertical worldline means x is constant: object at rest. A light ray has a 45° line if both axes are scaled equally.

Common mistake

Do not read a spacetime diagram like an ordinary map. The vertical axis is time. A diagonal line is not a spatial road. It is a history of motion.

7 The Light Clock and Time Dilation

7.1 The light clock setup

Imagine a clock made of two mirrors. A light pulse bounces between them. Each bounce is a tick. In the clock's own rest frame, the light travels straight up and down.

Let the distance between the mirrors be L. One half-tick takes time

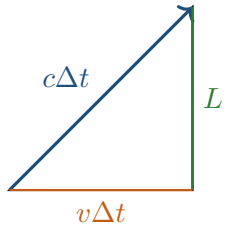
$$\Delta t_0 = \frac{L}{c}.$$

The full tick takes $2L/c$. The subscript 0 means **proper time**: time measured in the frame where the clock is at rest.

7.2 The same clock seen from another frame

Now suppose the clock moves horizontally at speed v. To an outside observer, the light follows a diagonal path. The light still moves at speed c, but it has to cover a longer path.

For half a tick:



Light clock geometry for half a tick

The right triangle gives

$$(c\Delta t)^2 = L^2 + (v\Delta t)^2.$$

Move the $v^2\Delta t^2$ term to the left:

$$c^2\Delta t^2 - v^2\Delta t^2 = L^2.$$

Factor:

$$\Delta t^2(c^2 - v^2) = L^2.$$

So

$$\Delta t = \frac{L}{\sqrt{c^2 - v^2}}.$$

Since $\Delta t_0 = L/c$,

$$\Delta t = \frac{\Delta t_0}{\sqrt{1 - v^2/c^2}}.$$

Define

$$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}}.$$

Then

$$\boxed{\Delta t = \gamma \Delta t_0.}$$

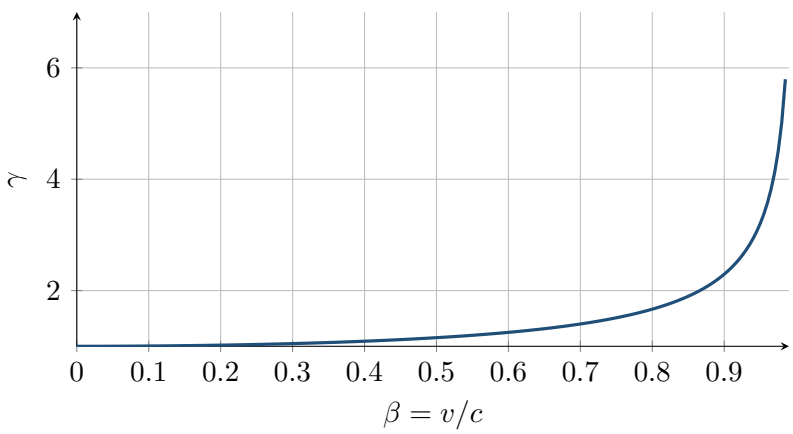
Key idea

A moving clock is measured to tick more slowly. The time interval between its ticks is longer by a factor γ.

7.3 Understanding gamma

The Lorentz factor is

$$\gamma = \frac{1}{\sqrt{1 - \beta^2}}, \qquad \beta = \frac{v}{c}.$$



For small speeds, γ is almost 1. That is why we do not notice relativity in daily life.

v/c	γ	Effect size
0.01	1.00005	invisible in daily life
0.10	1.005	small
0.50	1.155	noticeable
0.80	1.667	large
0.99	7.089	huge

Worked example

A spaceship moves at $v = 0.80c$. How much time passes on Earth if the astronaut's clock measures 3 years?
Here

$$\gamma = \frac{1}{\sqrt{1-0.8^2}} = \frac{1}{\sqrt{0.36}} = \frac{1}{0.6} = 1.667.$$

Therefore

$$\Delta t = \gamma \Delta t_0 = 1.667 \cdot 3 \approx 5.0 \text{ years}.$$

Earth measures about 5 years.

Common mistake

The moving clock is not broken. Time dilation is not a mechanical defect. It is a property of spacetime measurement.

8 Length Contraction

8.1 Proper length

The **proper length** L_0 of an object is the length measured in the frame where the object is at rest.

If the object moves relative to you at speed v, you measure its length along the direction of motion as

$$\boxed{L = \frac{L_0}{\gamma}.}$$

Moving objects are shorter along the direction of motion.

8.2 Why simultaneity matters

To measure the length of a moving object, you must record the positions of both ends at the same time in your frame. That phrase "at the same time" is not harmless. Different observers do not always agree about simultaneity.

Key idea

Length contraction is not a visual camera effect. It is a measurement effect involving simultaneous endpoint measurements in a chosen frame.

Worked example

A spaceship has proper length $L_0 = 100\,\text{m}$ and moves at $0.6c$. Find its measured length in the Earth frame.
First compute

$$\gamma = \frac{1}{\sqrt{1-0.6^2}} = \frac{1}{\sqrt{0.64}} = 1.25.$$

Then

$$L = \frac{L_0}{\gamma} = \frac{100}{1.25} = 80\,\text{m}.$$

Earth observers measure the spaceship length as $80\,\text{m}$.

9 Relativity of Simultaneity

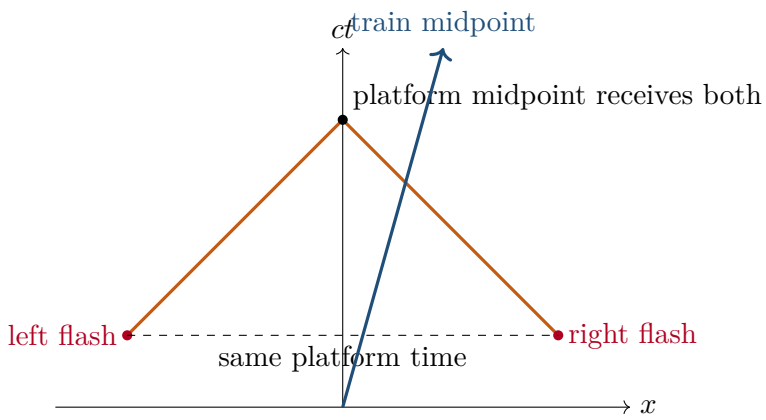
9.1 The train thought experiment

A train moves past a platform. Lightning strikes both ends of the train. A platform observer at the midpoint receives the two light signals at the same time and concludes: the strikes were simultaneous in the platform frame.

But a passenger at the midpoint of the train moves toward one light signal and away from the other. Since the passenger also measures light speed as c, the passenger concludes that one strike happened before the other.

Key idea

Events that are simultaneous in one inertial frame need not be simultaneous in another inertial frame.



9.2 The key lesson

This is usually the hardest conceptual step. Once simultaneity is relative, time dilation and length contraction become less paradoxical. They are different pieces of one spacetime geometry.

10 Lorentz Transformations

10.1 What transformations must do

We need equations replacing the Galilean transformations. They must satisfy:

1. If $v \ll c$, they should look like Galilean transformations.
2. They should make all inertial observers measure light speed c.
3. They should be linear for inertial frames, because uniform motion should map to uniform motion.

For motion along the x-axis:

$$\boxed{x' = \gamma(x - vt)},$$

$$\boxed{t' = \gamma\left(t - \frac{vx}{c^2}\right)}.$$

The perpendicular coordinates do not change:

$$y' = y, \qquad z' = z.$$

Common mistake

The time transformation contains x. This is the algebraic signal that time and space mix.

10.2 Check the low-speed limit

If $v \ll c$, then $\gamma \approx 1$ and vx/c^2 is tiny. Therefore

$$x' \approx x - vt, \qquad t' \approx t.$$

So Newtonian mechanics is recovered as an approximation.

10.3 Matrix form

Using coordinates (ct, x) and $\beta = v/c$, the Lorentz transformation can be written as

$$\begin{pmatrix} ct' \\ x' \end{pmatrix} = \begin{pmatrix} \gamma & -\gamma\beta \\ -\gamma\beta & \gamma \end{pmatrix} \begin{pmatrix} ct \\ x \end{pmatrix}.$$

This should look friendly to computer science students: it is a linear transformation, but not an ordinary Euclidean rotation.

Key idea

A Lorentz transformation is a change of inertial coordinates that preserves the spacetime interval.

10.4 A slow derivation of the Lorentz transformation

This derivation is not fully rigorous, but it shows why the formula has the shape it has.

Assume that S' moves at speed v relative to S. The origin of S' has coordinate $x' = 0$. In frame S, that origin moves as

$$x = vt.$$

So x' should contain the factor $x - vt$. We write

$$x' = A(x - vt),$$

where A is a factor to be found.

By symmetry, the inverse transformation should have the same form with v replaced by $-v$:

$$x = A(x' + vt').$$

Now impose the fact that light satisfies $x = ct$ in both frames. If a light pulse moves to the right, then

$$x = ct, \qquad x' = ct'.$$

Substitution and consistency force the factor

$$A = \gamma = \frac{1}{\sqrt{1 - v^2/c^2}}.$$

The corresponding time transformation is then

$$t' = \gamma \left(t - \frac{vx}{c^2} \right).$$

Key idea

The term vx/c^2 is small at ordinary speeds, but conceptually it is enormous: it means that time assigned to a distant event depends on the observer's state of motion.

10.5 Worked coordinate transformation

Worked example

An event has coordinates $t = 2.0\,\mu\text{s}$ and $x = 300\,\text{m}$ in frame S. Frame S' moves at $v = 0.5c$ along the x direction. Find t' and x'.
First compute

$$\gamma = \frac{1}{\sqrt{1 - 0.5^2}} = \frac{1}{\sqrt{0.75}} \approx 1.155.$$

Use $c = 3.00 \times 10^8\,\text{m/s}$ and $t = 2.0 \times 10^{-6}\,\text{s}$. Then

$$vt = (0.5)(3.00 \times 10^8)(2.0 \times 10^{-6}) = 300\,\text{m}.$$

So

$$x' = \gamma(x - vt) = 1.155(300 - 300) = 0.$$

For the time coordinate:

$$\frac{vx}{c^2} = \frac{(0.5c)(300)}{c^2} = \frac{150}{c} = \frac{150}{3.00 \times 10^8} = 0.5 \times 10^{-6}\,\text{s}.$$

Thus

$$t' = 1.155(2.0 - 0.5)\,\mu\text{s} \approx 1.73\,\mu\text{s}.$$

11 The Spacetime Interval

11.1 The invariant quantity

In Euclidean geometry, the distance between two points is

$$\Delta s^2 = \Delta x^2 + \Delta y^2 + \Delta z^2.$$

Rotating coordinates changes the coordinate components, but not the distance.

In special relativity, the invariant spacetime interval is

$$\boxed{\Delta s^2 = c^2\Delta t^2 - \Delta x^2 - \Delta y^2 - \Delta z^2.}$$

Different inertial observers may disagree on Δt and Δx, but they agree on Δs^2.

Common mistake

The minus signs are not decoration. Spacetime geometry is not ordinary Euclidean geometry with one extra dimension.

11.2 Types of intervals

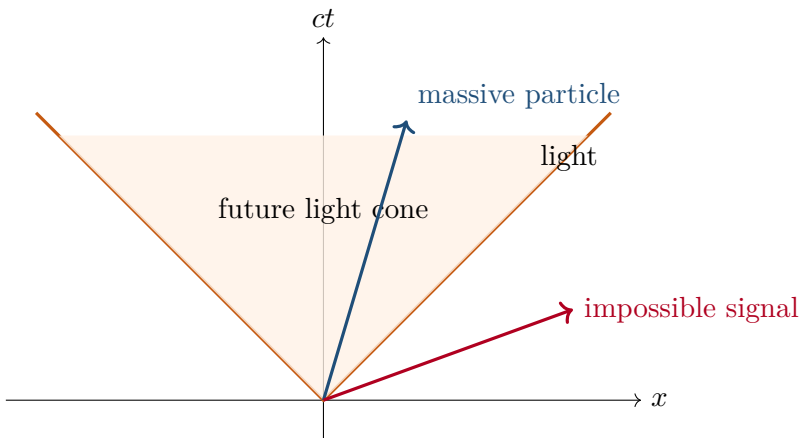
For simplicity, use one spatial dimension:

$$\Delta s^2 = c^2\Delta t^2 - \Delta x^2.$$

Type	Condition	Meaning
Timelike	$c^2\Delta t^2 > \Delta x^2$	one event can influence the other by slower-than-light motion
Lightlike	$c^2\Delta t^2 = \Delta x^2$	connected by light
Spacelike	$c^2\Delta t^2 < \Delta x^2$	no causal influence possible

11.3 Light cone

The light cone separates possible causal influence from impossible causal influence.



12 Causality and Information

12.1 Why faster-than-light signals are dangerous

If signals could travel faster than light, then in some frames the order of cause and effect could reverse. Relativity protects causality by separating spacetime into regions that can and cannot influence each other.

For computer science students, this resembles constraints in distributed systems: not all nodes can have global instantaneous agreement. Information propagation speed matters.

Key idea

Relativity makes causality geometric. Causal influence must stay inside or on the light cone.

12.2 No global universal clock

In distributed computing, clock synchronization is already hard because messages take time. Relativity says this is not merely an engineering inconvenience. At the fundamental level, there is no observer-independent global notion of simultaneous distant events.

13 Velocity Addition

13.1 Why ordinary addition fails

If a spaceship moves at $0.8c$ and fires a probe forward at $0.7c$ relative to the spaceship, Newtonian addition gives

$$0.8c + 0.7c = 1.5c.$$

This is impossible for a massive object.

The relativistic velocity addition formula is

$$\boxed{u = \frac{u' + v}{1 + \frac{u'v}{c^2}}.}$$

13.2 Worked example

Worked example

A spaceship moves at $v = 0.8c$ relative to Earth. It fires a probe forward at $u' = 0.7c$ relative to the spaceship. What speed does Earth measure?
Use

$$u = \frac{u' + v}{1 + u'v/c^2}.$$

Substitute:

$$u = \frac{0.7c + 0.8c}{1 + (0.7)(0.8)} = \frac{1.5c}{1.56} \approx 0.962c.$$

The result is below c.

Common mistake

You may add small velocities normally only when speeds are much smaller than c. At relativistic speeds, use the relativistic formula.

14 Proper Time

14.1 Time measured along a worldline

Proper time is the time measured by a clock traveling along a particular worldline. For small intervals in flat spacetime,

$$\mathrm{d}\tau = \mathrm{d}t\sqrt{1 - \frac{v^2}{c^2}}.$$

So

$$\Delta\tau = \int \sqrt{1 - \frac{v(t)^2}{c^2}}\,\mathrm{d}t.$$

If v is constant,

$$\Delta\tau = \frac{\Delta t}{\gamma}.$$

Key idea

Proper time is like the "length" of a path through spacetime, but with Minkowski geometry rather than Euclidean geometry.

14.2 Algorithmic viewpoint

If we have a numerical trajectory sampled in time, we can approximate proper time by summing small pieces.

```
import math

def proper_time(times, speeds, c=299792458.0):
    total = 0.0
    for i in range(len(times) - 1):
        dt = times[i+1] - times[i]
```

```
        v = speeds[i]
        total += dt * math.sqrt(1.0 - (v/c)**2)
    return total
```

This simple code is conceptually powerful: relativity is not only philosophy. It gives computable rules.

15 The Twin Paradox

15.1 The story

One twin stays on Earth. The other travels in a spaceship at high speed, turns around, and returns. The traveling twin is younger.

This sounds paradoxical because each twin might say that the other one was moving. But the situations are not symmetric: the traveling twin changes inertial frames during the turnaround.

15.2 Simple calculation

Worked example

A spaceship travels away from Earth at $0.8c$ for 5 years in the Earth frame, then returns at $0.8c$ for 5 years in the Earth frame.
Earth time:

$$\Delta t = 10 \text{ years}.$$

For $0.8c$,

$$\gamma = 1.667.$$

Traveler proper time:

$$\Delta\tau = \frac{10}{1.667} \approx 6.0 \text{ years}.$$

The astronaut ages about 6 years while Earth ages 10 years.

Common mistake

The twin paradox is not solved by saying "time dilation is only apparent". It is real: the twins can reunite and compare clocks.

16 Relativistic Momentum and Energy

16.1 Momentum

Classical momentum is

$$p = mv.$$

In special relativity, momentum becomes

$$\boxed{p = \gamma mv.}$$

As v approaches c, γ grows. It becomes harder and harder to increase the speed of a massive object.

16.2 Energy

The total relativistic energy is

$$\boxed{E = \gamma mc^2.}$$

When the object is at rest, $v = 0$ and $\gamma = 1$, so

$$\boxed{E_0 = mc^2.}$$

This is rest energy.

The kinetic energy is

$$K = E - E_0 = (\gamma - 1)mc^2.$$

16.3 Energy-momentum relation

The invariant relation is

$$\boxed{E^2 = p^2c^2 + m^2c^4.}$$

For a massless particle such as a photon, $m = 0$, so

$$E = pc.$$

Worked example

A particle has rest mass m and moves at $0.6c$. Find its total energy in units of mc^2 and kinetic energy in units of mc^2.
First:

$$\gamma = \frac{1}{\sqrt{1 - 0.6^2}} = 1.25.$$

Total energy:

$$E = \gamma mc^2 = 1.25mc^2.$$

Kinetic energy:

$$K = (\gamma - 1)mc^2 = 0.25mc^2.$$

17 Relativistic Doppler Effect

17.1 Why frequency changes

If a light source moves toward an observer, the observer measures a higher frequency. If it moves away, the observer measures a lower frequency. This is the Doppler effect. In relativity, time dilation also contributes, so the formula differs from the classical sound-wave formula.

For motion directly toward or away from the observer, the relativistic Doppler factor is

$$\frac{f_{\text{obs}}}{f_{\text{src}}} = \sqrt{\frac{1+\beta}{1-\beta}}$$

for approaching motion, and

$$\frac{f_{\text{obs}}}{f_{\text{src}}} = \sqrt{\frac{1-\beta}{1+\beta}}$$

for receding motion.

Worked example

A source moves away at $0.6c$. What fraction of the original frequency does the observer measure?
Use

$$\frac{f_{\text{obs}}}{f_{\text{src}}} = \sqrt{\frac{1-0.6}{1+0.6}} = \sqrt{\frac{0.4}{1.6}} = \sqrt{0.25} = 0.5.$$

The observed frequency is half the source frequency.

17.2 Redshift

Astronomers often describe the change by redshift z:

$$1 + z = \frac{\lambda_{\text{obs}}}{\lambda_{\text{emit}}}.$$

For light, lower frequency means longer wavelength. A receding source is redshifted.

Common mistake

Doppler redshift, gravitational redshift, and cosmological redshift are related ideas but not exactly the same physical mechanism.

18 Photons and Massless Particles

A photon has zero rest mass but nonzero energy and momentum. For a photon,

$$E = hf, \qquad p = \frac{E}{c} = \frac{hf}{c}.$$

This is consistent with the energy-momentum relation

$$E^2 = p^2c^2 + m^2c^4.$$

If $m = 0$, then

$$E^2 = p^2c^2, \qquad E = pc.$$

Key idea

Massless particles do not have a rest frame. You cannot ride alongside a photon at rest.

19 Four-Vectors

19.1 Why four-vectors matter

A vector in ordinary 3D space has components that change when we rotate coordinates. But the vector itself is a geometric object. In relativity, the same idea applies in spacetime.

A spacetime position four-vector is

$$X^\mu = (ct, x, y, z).$$

The Greek index μ usually runs over $0, 1, 2, 3$.

Definition

A **four-vector** is an object with four components that transforms under Lorentz transformations in the same way as spacetime coordinates.

19.2 Minkowski dot product

For two four-vectors $A^\mu = (A^0, A^1, A^2, A^3)$ and $B^\mu = (B^0, B^1, B^2, B^3)$, a common sign convention gives

$$A \cdot B = A^0 B^0 - A^1 B^1 - A^2 B^2 - A^3 B^3.$$

This dot product is invariant under Lorentz transformations.

19.3 Four-momentum

The energy and momentum combine into a four-vector:

$$P^\mu = \left(\frac{E}{c}, p_x, p_y, p_z \right).$$

Its invariant norm gives

$$P \cdot P = m^2 c^2.$$

This is equivalent to

$$E^2 = p^2 c^2 + m^2 c^4.$$

20 Special Relativity as Geometry

Special relativity becomes simpler when viewed as geometry.

Euclidean geometry	Relativistic spacetime	Invariant
Rotations	Lorentz transformations	interval
Distance	Spacetime interval	$c^2 \Delta t^2 - \Delta x^2 - \cdots$
Vectors	Four-vectors	Minkowski norm
Angle-like parameter	Rapidity	velocity composition

20.1 Rapidity

Velocity can be parameterized by rapidity ϕ:

$$\beta = \tanh \phi, \qquad \gamma = \cosh \phi, \qquad \gamma\beta = \sinh \phi.$$

Why is this useful? Rapidities add simply. If one boost has rapidity ϕ_1 and another has rapidity ϕ_2, the combined boost has rapidity

$$\phi_1 + \phi_2.$$

This is like using logarithms to turn multiplication into addition.

21 Acceleration in Relativity

21.1 Coordinate acceleration versus proper acceleration

Acceleration is more subtle in relativity. Coordinate acceleration depends on the observer's frame. Proper acceleration is what an accelerometer carried by the object measures.

A spaceship can feel constant proper acceleration, even though its coordinate speed approaches c but never reaches it.

Key idea

Relativity does not forbid acceleration. It forbids accelerating a massive object through the speed of light.

21.2 Why massive objects cannot reach light speed

As $v \to c$, the Lorentz factor

$$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}}$$

goes to infinity. Therefore the energy

$$E = \gamma mc^2$$

also goes to infinity for $m > 0$.

No finite energy input can accelerate a massive object to c.

22 From Special to General Relativity

Special relativity describes inertial frames in flat spacetime. But gravity creates a problem. Newtonian gravity acts instantaneously at a distance. Special relativity says causal influence cannot travel faster than light.

Einstein's next step was to rethink gravity itself.

Key idea

General relativity is the theory in which gravity is not treated as an ordinary force. It is described as curvature of spacetime.

23 The Equivalence Principle

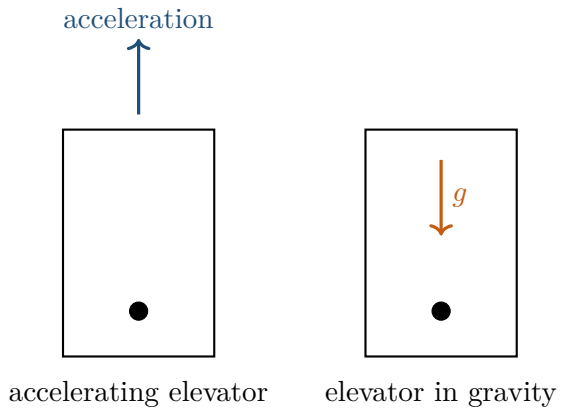
23.1 Elevator idea

Imagine you are inside a closed elevator. If the elevator is in deep space accelerating upward at g, you feel pressed to the floor. If the elevator is standing on Earth, you also feel pressed to the floor.

Locally, these two situations are physically similar.

Definition

Equivalence principle: Locally, the effects of a uniform gravitational field are indistinguishable from the effects of acceleration.



23.2 A very important consequence

If acceleration can bend the apparent path of light, then gravity should bend light too. This was one of the key predictions of general relativity.

24 Gravity as Curvature

24.1 What curvature means

Curvature does not mean that spacetime is embedded in some higher-dimensional space like a rubber sheet. The rubber sheet picture is useful but dangerous. Curvature means that the rules of geometry inside spacetime are not globally Euclidean/Minkowskian.

Examples of curvature:

- The sum of angles in a large triangle may differ from 180°.
- Initially parallel geodesics may converge or diverge.
- Clocks at different gravitational potentials tick at different rates.

Common mistake

The common rubber sheet image is only an analogy. Real spacetime curvature includes time, not just a curved spatial surface.

24.2 Geodesics

In ordinary Euclidean space, a free particle moves along a straight line. In curved spacetime, a free-falling particle follows a geodesic.

Definition

A **geodesic** is the natural straightest possible path in a curved geometry. In general relativity, freely falling objects follow geodesics in spacetime.

Key idea

Gravity is not a force pulling a freely falling object away from its natural path. Free fall is the natural path. The ground pushing on you is what prevents you from following a geodesic.

25 The Metric Tensor

25.1 From intervals to metrics

In special relativity, the interval is

$$\mathrm{d}s^2 = c^2\mathrm{d}t^2 - \mathrm{d}x^2 - \mathrm{d}y^2 - \mathrm{d}z^2.$$

This can be written using a metric tensor:

$$\mathrm{d}s^2 = g_{\mu\nu}\mathrm{d}x^\mu \mathrm{d}x^\nu.$$

For flat spacetime in Cartesian coordinates,

$$(g_{\mu\nu}) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

if we use coordinates (ct, x, y, z).

25.2 What the metric does

The metric tells us how to compute spacetime intervals from coordinate differences. In curved spacetime, the components $g_{\mu\nu}$ can depend on position.

Key idea

For a computer scientist, the metric is like a local rule used by a function that computes lengths, times, and angles from coordinate data.

26 Gravitational Time Dilation

Clocks deeper in a gravitational field tick more slowly compared with clocks far away.

Near a spherical mass, the Schwarzschild metric gives approximately

$$\Delta t_{\text{far}} = \frac{\Delta\tau_{\text{near}}}{\sqrt{1 - \frac{2GM}{rc^2}}}.$$

Equivalently,

$$\Delta\tau_{\text{near}} = \Delta t_{\text{far}}\sqrt{1 - \frac{2GM}{rc^2}}.$$

Worked example

Suppose a clock is closer to a massive body. The factor

$$\sqrt{1 - \frac{2GM}{rc^2}}$$

is less than 1. Therefore the proper time near the mass is smaller than the far-away coordinate time. The near clock ticks slower relative to far-away observers.

26.1 GPS connection

GPS satellites require relativistic corrections. Their clocks are affected by both special relativistic time dilation due to motion and general relativistic time dilation due to weaker gravity at orbital altitude. Without corrections, positioning errors would grow rapidly.

This is an important practical lesson: relativity is not only astronomy. It is part of real infrastructure.

27 Geodesics as an Algorithmic Problem

The full geodesic equation is

$$\frac{\mathrm{d}^2 x^\mu}{\mathrm{d}\lambda^2} + \Gamma^\mu_{\alpha\beta} \frac{\mathrm{d}x^\alpha}{\mathrm{d}\lambda} \frac{\mathrm{d}x^\beta}{\mathrm{d}\lambda} = 0.$$

Do not panic. At this level, the equation says:

- $x^\mu(\lambda)$ describes a path through spacetime.
- λ is a parameter along the path.
- $\Gamma^\mu_{\alpha\beta}$ are Christoffel symbols, computed from the metric.
- The metric determines the path followed by free particles and light.

For a programmer, this resembles a system of ordinary differential equations. Given initial position and velocity, integrate forward.

```
# Pseudocode only: symbolic details omitted
state = initial_position_and_velocity
for step in range(n_steps):
    acceleration = geodesic_acceleration(metric, state)
    state = integrator_step(state, acceleration, dt)
```

Common mistake

The hard part in numerical relativity is not writing a loop. The hard part is using the correct geometry, stable coordinates, constraints, and numerical method.

28 Weak-Field Gravity and Newtonian Limit

General relativity must reproduce Newtonian gravity when fields are weak and speeds are small. In that limit, the time-time part of the metric is related to the Newtonian gravitational potential

Φ approximately by

$$g_{00} \approx 1 + \frac{2\Phi}{c^2}.$$

This is one reason the metric is physically meaningful: it contains gravitational information.

Key idea

Newtonian gravity is the low-speed, weak-field approximation of general relativity.

Worked example

Near Earth, gravitational effects on time are tiny but measurable. If two clocks are placed at different heights, the higher clock is slightly farther from Earth's mass and ticks slightly faster. This is not science fiction. Precision clocks can measure it.

29 Einstein Field Equations: Meaning, Not Full Derivation

The field equations of general relativity are

$$\boxed{G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}.}$$

At beginner level, read this as:

Symbol	Rough meaning	Role
$G_{\mu\nu}$	curvature of spacetime	geometry side
$T_{\mu\nu}$	energy, momentum, pressure, stress	matter side
G	gravitational constant	coupling strength
c	speed of light	relativistic scale

Key idea

Matter and energy tell spacetime how to curve. Curved spacetime tells matter and light how to move.

30 Schwarzschild Geometry and Black Holes

30.1 Schwarzschild radius

For a non-rotating spherical mass M, the Schwarzschild radius is

$$\boxed{r_s = \frac{2GM}{c^2}.}$$

If mass is compressed inside this radius, a black hole forms.

Worked example

Estimate the Schwarzschild radius of Earth.

Use:

$$G = 6.67 \times 10^{-11}\,\mathrm{N\,m^2/kg^2}, \quad M_E = 5.97 \times 10^{24}\,\mathrm{kg}, \quad c = 3.00 \times 10^{8}\,\mathrm{m/s}.$$

Then

$$r_s = \frac{2GM_E}{c^2} \approx \frac{2(6.67 \times 10^{-11})(5.97 \times 10^{24})}{(3.00 \times 10^{8})^2}.$$

Compute numerator:

$$2(6.67)(5.97) \times 10^{13} \approx 79.6 \times 10^{13} = 7.96 \times 10^{14}.$$

Denominator:

$$9.00 \times 10^{16}.$$

So

$$r_s \approx 8.84 \times 10^{-3}\,\mathrm{m}.$$

Earth's Schwarzschild radius is about 9 mm.

30.2 Event horizon

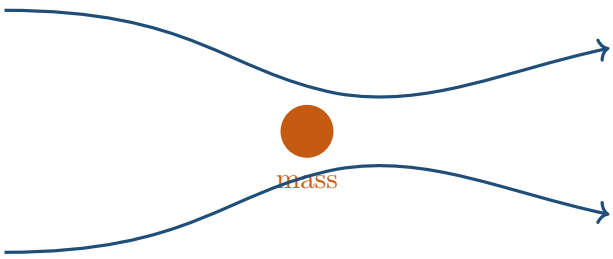
The event horizon is not a material surface. It is a causal boundary. Once inside, future-directed paths cannot escape to infinity.

Common mistake

A black hole is not a cosmic vacuum cleaner. Far away from it, its gravity acts like the gravity of any object with the same mass.

31 Light Bending and Gravitational Lensing

Light follows null geodesics. In curved spacetime, these paths can bend around massive objects. This produces gravitational lensing.



Light paths bending near a massive object

32 Cosmology in One Lecture

General relativity also describes the universe as a whole. If the universe is homogeneous and isotropic on large scales, its geometry can be described by a scale factor $a(t)$.

Distances between far-away galaxies grow roughly as the scale factor grows. This is the expansion of the universe.

Common mistake

Galaxies are not simply flying through pre-existing space from a central explosion. In modern cosmology, space itself expands on large scales.

The Hubble law for nearby galaxies is

$$v \approx H_0 d.$$

It says that recession speed is approximately proportional to distance.

33 Relativity and Computation

33.1 Numerical simulations

Relativity is full of computable structures:

- Lorentz transformations are matrix operations.
- Worldlines can be sampled and integrated numerically.
- Proper time can be approximated by sums.
- Geodesics can be solved as differential equations.
- Relativistic ray tracing is used for visualizing black holes.

33.2 Simple Lorentz transformation function

```
import math

c = 299792458.0

def lorentz_transform(t, x, v):
    beta = v / c
    gamma = 1.0 / math.sqrt(1.0 - beta*beta)
    t_prime = gamma * (t - v*x/(c*c))
    x_prime = gamma * (x - v*t)
    return t_prime, x_prime
```

This code encodes one of the central formulas of special relativity.

33.3 A unit-friendly version

Physicists often set $c = 1$. Then formulas become simpler:

$$x' = \gamma(x - vt), \qquad t' = \gamma(t - vx),$$

where v is now measured as a fraction of c.

Common mistake

Setting $c = 1$ is convenient, but it can confuse beginners. Always know whether v is in meters per second or in units of c.

34 Step-by-Step Study Plan

If a student feels lost, use this order:

1. Learn what an event is.
2. Learn to draw a spacetime diagram.
3. Understand that light has the same speed for all inertial observers.
4. Derive time dilation with the light clock.
5. Learn length contraction only after understanding simultaneity.
6. Use Lorentz transformations for calculations.
7. Use the spacetime interval to classify causal relations.
8. Learn energy and momentum after the geometry makes sense.
9. Treat general relativity first as geometry, not as advanced tensor calculus.

Key idea

The safest learning path is: events → diagrams → invariants → transformations → applications.

35 More Worked Examples for Weak Students

35.1 Example A: deciding whether relativity matters

Worked example

A car moves at 30 m/s. Is special relativity important for ordinary calculations?
Compute

$$\beta = \frac{30}{3.00 \times 10^8} = 1.0 \times 10^{-7}.$$

Then

$$\beta^2 = 10^{-14}.$$

So

$$\gamma = \frac{1}{\sqrt{1 - 10^{-14}}} \approx 1.$$

Relativistic corrections are completely negligible for normal car motion.

35.2 Example B: avoiding a unit trap

Worked example

A signal travels for 4 ns. How far does light travel in this time?
A nanosecond is

$$1\,\text{ns} = 10^{-9}\,\text{s}.$$

So

$$4\,\text{ns} = 4 \times 10^{-9}\,\text{s}.$$

The distance is

$$d = ct = (3.00 \times 10^8)(4 \times 10^{-9}) = 1.2\,\text{m}.$$

Light travels about 1.2 m in 4 ns.

35.3 Example C: checking causality

Worked example

Event A happens at $(t, x) = (0, 0)$. Event B happens at $t = 1\,\mu\text{s}$ and $x = 600\,\text{m}$. Can A influence B?
Light travels in $1\,\mu\text{s}$ a distance

$$c\Delta t = (3.00 \times 10^8)(10^{-6}) = 300\,\text{m}.$$

But B is 600 m away. Since $600 > 300$, even light cannot get there in time. The separation is spacelike, so A cannot influence B.

36 Typical Exam Problems

36.1 Problem 1: time dilation

A particle lives for $2.0\,\mu\text{s}$ in its own rest frame. In a laboratory it moves at $0.98c$. How long does it live in the laboratory frame?

Solution.

$$\gamma = \frac{1}{\sqrt{1 - 0.98^2}} = \frac{1}{\sqrt{1 - 0.9604}} = \frac{1}{\sqrt{0.0396}} \approx 5.03.$$

Therefore

$$\Delta t = \gamma \Delta t_0 = 5.03 \cdot 2.0\,\mu\text{s} \approx 10.1\,\mu\text{s}.$$

36.2 Problem 2: length contraction

A rod has proper length 4.0 m. It moves past an observer at $0.75c$. What length does the observer measure?

Solution.

$$\gamma = \frac{1}{\sqrt{1 - 0.75^2}} = \frac{1}{\sqrt{0.4375}} \approx 1.51.$$

Thus

$$L = \frac{L_0}{\gamma} = \frac{4.0}{1.51} \approx 2.65\,\text{m}.$$

36.3 Problem 3: interval classification

Two events have $\Delta t = 5\,\text{s}$ and $\Delta x = 2c\,\text{s}$ in one dimension. Classify the interval.

Solution. Here $\Delta x = 2c\,\text{s}$ means the spatial separation equals the distance light travels in 2 seconds. Compute:

$$c^2\Delta t^2 - \Delta x^2 = c^2(5^2) - c^2(2^2) = 21c^2 > 0.$$

The interval is timelike.

36.4 Problem 4: relativistic velocity addition

A ship moves at $0.6c$ relative to Earth and emits a probe forward at $0.6c$ relative to the ship. Find the probe speed relative to Earth.

Solution.

$$u = \frac{0.6c + 0.6c}{1 + 0.6 \cdot 0.6} = \frac{1.2c}{1.36} \approx 0.882c.$$

36.5 Problem 5: rest energy

Find the rest energy of 1 kg of mass.

Solution.

$$E_0 = mc^2 = 1 \cdot (3.00 \times 10^8)^2 = 9.00 \times 10^{16}\,\text{J}.$$

This is enormous.

37 Common Misconceptions

1. **"Relativity says everything is relative."** No. It says some quantities are frame-dependent and others are invariant.
2. **"Time dilation is an illusion."** No. Reunited clocks can show different elapsed times.
3. **"Mass increases with speed."** Modern treatments usually keep rest mass invariant and say energy and momentum increase.
4. **"Nothing can move faster than light."** More precisely: no material object or information signal can be accelerated through or transmitted faster than c locally.
5. **"Gravity is just a force in general relativity."** In GR, free fall is geodesic motion in curved spacetime.
6. **"Black holes suck everything in."** Far away, they behave gravitationally like any mass of the same value.

38 Formula Sheet

Special relativity

$$\beta = \frac{v}{c}$$

$$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}}$$

$$\Delta t = \gamma \Delta t_0$$

$$L = \frac{L_0}{\gamma}$$

$$x' = \gamma(x - vt)$$

$$t' = \gamma\left(t - \frac{vx}{c^2}\right)$$

$$\Delta s^2 = c^2\Delta t^2 - \Delta x^2 - \Delta y^2 - \Delta z^2$$

$$u = \frac{u' + v}{1 + u'v/c^2}$$

Energy and momentum

$$p = \gamma m v$$

$$E = \gamma m c^2$$

$$E_0 = mc^2$$
$$K = (\gamma - 1)mc^2$$
$$E^2 = p^2c^2 + m^2c^4$$

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

$$r_s = \frac{2GM}{c^2}$$

General relativity

$$\mathrm{d}s^2 = g_{\mu\nu}\mathrm{d}x^\mu \mathrm{d}x^\nu$$

$$\Delta\tau = \Delta t\sqrt{1 - \frac{2GM}{rc^2}}$$

39 Exercises

Practice

Exercise 1. Compute γ for $v = 0.3c$, $0.6c$, and $0.9c$.

Practice

Exercise 2. A spaceship moves at $0.9c$. Its onboard clock measures 4 years. How much time passes in the Earth frame?

Practice

Exercise 3. A rod has proper length 12 m and moves at $0.8c$. What is its contracted length?

Practice

Exercise 4. Classify an interval with $\Delta t = 3\,\mathrm{s}$ and $\Delta x = 5c\,\mathrm{s}$.

Practice

Exercise 5. A ship moving at $0.5c$ fires a probe backward at $0.5c$ relative to the ship. What speed does Earth measure?

Practice

Exercise 6. Write a function that accepts t, x, v and returns t', x' using the Lorentz transformation. Test it for $v = 0$.

40 Selected Answers

Exercise 1.

$$\gamma(0.3c) \approx 1.048, \quad \gamma(0.6c) = 1.25, \quad \gamma(0.9c) \approx 2.294.$$

Exercise 2. For $0.9c$, $\gamma \approx 2.294$, so

$$\Delta t \approx 2.294 \cdot 4 = 9.18 \text{ years}.$$

Exercise 3. For $0.8c$, $\gamma = 1.667$, so

$$L = 12/1.667 \approx 7.2\,\mathrm{m}.$$

Exercise 4.

$$c^2\Delta t^2 - \Delta x^2 = c^2(9) - c^2(25) < 0.$$

The interval is spacelike.

Exercise 5. Use $u' = -0.5c$, $v = 0.5c$:

$$u = \frac{-0.5c + 0.5c}{1 - 0.25} = 0.$$

Earth measures the probe as momentarily at rest.

41 Final Summary

Special relativity begins with a simple but radical claim: all inertial observers measure the same speed of light. From that claim follow time dilation, length contraction, relativity of simultaneity, Lorentz transformations, and the invariant spacetime interval.

General relativity goes further. It says that gravity is not an ordinary force acting instantly across space. Gravity is the geometry of spacetime, and free-falling objects follow geodesics in that geometry.

For beginners, the most important habit is this:

Key idea

Do not ask only "what is the formula?" Ask: which observer measures this quantity, and which quantity remains invariant when the observer changes?